

# Reinforcement Learning using EEG signals for Therapeutic Use of Music in Emotion Management

Tirth Patel\*, Sandeep Eslavath\*, Het Talavia\*

\*Department of Artificial Intelligence

Sardar Vallabhbhai National Institute of Technology (SVNIT), Surat, India

{u23ai072, u23ai081, u23ai077}@coed.svnit.ac.in

**Abstract**—Music therapy is becoming a popular way to improve our mental and emotional well-being without relying on medication. But most music recommendation systems focus on what we like, rather than how we’re feeling at the moment. To address this, we’ve developed a system that uses music to help regulate our emotions. We’ve modeled this as a problem of making sequential decisions to move our emotional state in a positive direction. Our system uses a type of artificial intelligence called Q-learning to guide these emotional transitions. We’ve built a Deep Q-Network (DQN) model that selects a series of short music clips to help move us from a negative emotional state, like anger, to a more positive one, like happiness, in just a few steps. We’ve tested our model using a dataset called DEAP and evaluated its performance across 32 subjects. We compared our DQN model to a simpler baseline model, as well as three other variations: one that incorporates emotional velocity features, one that uses time-dependent reward shaping, and one that navigates across multiple emotional targets. Our continuous DQN model performed the best, achieving a 93.8 success rate in moving subjects to their target emotional state, with an average error of just 5 degrees. This represents a significant improvement over the baseline model, with a 25 percentage-point increase in success rate and a 3.2 times reduction in error. Our results also suggest that there’s an imbalance in the DEAP dataset, with positive emotional targets being easier to reach than negative ones. Overall, our findings indicate that using continuous state representations and subject-independent action abstractions can be effective for managing emotions through music. This has important implications for the development of music therapy systems that can help people improve their mental and emotional well-being. We’re excited about the potential of music therapy to make a positive impact on people’s lives, and we believe that our system can be a valuable tool in this effort. By using music to regulate our emotions, we can take a more proactive approach to managing our mental health and well-being. And with the help of artificial intelligence, we can develop systems that are tailored to our individual needs and preferences, making music therapy more effective and accessible than ever before. In the future, we hope to continue refining our system and exploring new ways to use music therapy to improve people’s lives. We believe that music has the power to transform and uplift us, and we’re committed to harnessing that power to make a positive difference in the world. Whether you’re struggling with anxiety, depression, or simply looking for a way to improve your overall well-being, music therapy can be a powerful tool to help you achieve your goals. So why not give it a try? You might be surprised at the positive impact it can have on your life.

## I. INTRODUCTION

Hundreds of millions of people are affected by depression across worldwide [1], which increases the need for accessible and low-risk treatments such as music therapy. However,

most music recommendation systems are designed to maximize user preference and engagement rather than improve emotional well-being. Accordingly, an emotion-management system must focus on *how a sequence of clips changes affect over time*, not only on which individual clip which user might prefer to listen.

When we think about music that helps us feel better, we can look at it like a step-by-step process. Each piece of music we choose can change how we’re feeling at that moment. In this study, we use special equipment to measure how someone is feeling, using things like how happy or sad they are, and how excited or calm they are. Then, we use this information to pick a series of songs that can help the person get from how they’re feeling now to how they want to feel. It’s like creating a personalized playlist to help improve your mood.

Dutta et al. [2] took an early step in this direction by applying tabular Q-learning to the DEAP dataset [3]. Their method discretises the valence–arousal (V–A) plane into nine regions using the Geneva Emotion Wheel (GEW) [5]. While effective as a baseline, this design has two limitations: (1) a *state bottleneck*, where continuous affect is collapsed into a small number of bins; and (2) an *action bottleneck*, where treating each of the 40 clips as a distinct action hinders cross-subject transfer.

We address these limitations through four design choices:

- **INN-1:** A continuous DQN that directly processes raw  $(v, a)$  coordinates, eliminating discretisation loss.
- **INN-2:** A momentum-augmented DQN with 4-D state  $(v, a, \Delta v, \Delta a)$  capturing emotional trajectory.
- **INN-3:** Adaptive curriculum-style reward shaping with time-dependent penalty scaling.
- **INN-4:** Multi-target navigation evaluating all eight GEW directions to expose dataset biases.

The continuous DQN, also known as INN-1, does really well in the LOSO evaluation - it gets to 93.8% convergence with 32 subjects. This is a big deal because it also cuts down the mean angular error by a factor of 3.2 compared to the baseline.

## II. METHODOLOGY

### A. Data Preprocessing

We use self-report valence and arousal ratings from DEAP (32 subjects  $\times$  40 clips). Since rating scales differ across participants, we apply a two-stage normalisation:

1) *Z-Score Standardisation*: For each subject  $s$  and clip  $i$ , we compute the z-score using the mean and standard deviation of subject  $s$ 's ratings across all 40 clips.

$$z_i^{(s)} = \frac{x_i^{(s)} - \mu^{(s)}}{\sigma^{(s)}} \quad (1)$$

where  $\mu^{(s)}$  and  $\sigma^{(s)}$  are the mean and standard deviation of subject  $s$ 's ratings across all 40 clips.

2) *Hyperbolic Tangent Bounding*: We use the hyperbolic tangent function to keep the normalized values between -1 and 1. This is applied to both the valence and arousal z-scores.

$$v_i, a_i = \tanh(z_{i,\text{valence}}), \tanh(z_{i,\text{arousal}}) \quad (2)$$

By doing this, we create a continuous state space, which we call  $\mathcal{S}$ , that is contained within a square area where both  $x$  and  $y$  values are between -1 and 1, i.e.,  $\mathcal{S} \subseteq [-1, 1]^2$ .

### B. MDP Formulation

We set up this task of navigating emotions like a problem that can be solved step by step, using something called a Markov Decision Process, or MDP  $\langle \mathcal{S}, \mathcal{A}, \mathcal{T}, \mathcal{R}, \gamma \rangle$  for short.

- **State Space  $\mathcal{S}$** : To start, we break down the emotions into 9 simple groups - 8 main feelings and neutral. We do this by looking at how similar they are. But for the new way of learning, we use a continuous state: it's made up of two things, valence and arousal, and each of these can be any number between -1 and 1, i.e.,  $\mathbf{s} = (v, a) \in [-1, 1]^2$ . We're changing the original 40 actions that were specific to each person, and instead using 8 more general directions that can work for anyone, which allows us to use the same policy for different people.
- **Transition  $\mathcal{T}$** : Applying action  $a$  in state  $\mathbf{s}$  selects a music clip via weighted cosine similarity (Eq. 4) and transitions to the clip's V-A coordinates.
- **Reward  $\mathcal{R}$** : Binary reward with shaping penalty for stagnation (detailed in Section II-C).
- **Discount  $\gamma$** : Standard temporal discounting.

1) *State Quantisation (Baseline)*: So we have a state, which is made up of two things: velocity and acceleration, and we're trying to figure out which group it belongs to. Given a continuous state  $\mathbf{s} = (v, a)$  and GEW centroids  $C = \{\mathbf{c}_1, \dots, \mathbf{c}_8\}$ , we use something called cosine similarity to decide this:

$$S = \arg \max_{c \in C} \frac{\mathbf{s} \cdot \mathbf{c}}{\|\mathbf{s}\| \|\mathbf{c}\|} \quad (3)$$

Basically, we look at how similar the state is to each of the groups, and we assign it to the one that's most similar. But if the state isn't very strong, meaning its magnitude is below a certain threshold, then we just call it Neutral. This threshold is like a cutoff point, and it's represented by the symbol  $\epsilon$ , i.e., if  $\|\mathbf{s}\| < \epsilon$ , the state is classified as Neutral.

2) *Action-to-Clip Resolution*: When an agent picks a general direction for music, we figure out which specific music clip to use by combining two things: how similar it is to the main goal, and how similar it is to what's happening now.

$$i^* = \arg \max_{i \in \{1..40\}} [0.7 \cdot \text{sim}(\mathbf{c}_i, \mathbf{c}_{\text{target}}) + 0.3 \cdot \text{sim}(\mathbf{c}_i, \mathbf{s}_t)] \quad (4)$$

We give more importance to the main goal, about 70%, and less to what's happening now, about 30%, where  $\text{sim}(\cdot, \cdot)$  denotes cosine similarity.

### C. Reward Function

The reward is simple - the agent gets a point if it reaches the target area, and nothing if it doesn't, i.e.,  $r = 1$  if the agent reaches the target region,  $r = 0$  otherwise. But to make things more interesting, we're adding a penalty to deal with the fact that the rewards are pretty sparse. This penalty is based on an idea called potential-based shaping [7]:

$$R'_{t+1} = R_{t+1} + \begin{cases} \lambda & \text{if } S_{t+1} = S_t \text{ and } S_{t+1} \neq S_{\text{target}} \\ 0 & \text{otherwise} \end{cases} \quad (5)$$

Basically, if the agent gets stuck in the same spot that's not the target, it gets a penalty of  $\lambda = -100$ . This penalty is like a reminder to the agent to keep moving and not get too comfortable in one place.

### D. Baseline: Tabular Q-Learning

The baseline agent maintains a Q-table of size  $|\mathcal{S}| \times |\mathcal{A}|$  updated via the Bellman equation:

$$Q(S_t, A_t) \leftarrow Q(S_t, A_t) + \alpha [R_{t+1} + \gamma \max_a Q(S_{t+1}, a) - Q(S_t, A_t)] \quad (6)$$

Training runs for 1,000 epochs per LOSO fold.

### E. INN-1: Continuous Deep Q-Network

We're using a 3-layer multilayer perceptron, or MLP, to replace the Q-table:

$$\mathbf{s} = (v, a) \xrightarrow{\text{Dense}(48)+\text{ReLU}} \xrightarrow{\text{Dense}(48)+\text{ReLU}} \xrightarrow{\text{Dense}(8)} \mathbf{Q}(\mathbf{s}, \cdot) \quad (7)$$

This MLP has two hidden layers, each with 48 units, and it uses ReLU activation. The output layer has 8 units. Our goal is to train the DQN to minimize the MSE loss:

$$L(\theta) = \mathbb{E}_{(s,a,r,s') \sim D} [(y_j - Q(s_j, a_j; \theta))^2] \quad (8)$$

and we're doing this with a separate target network  $\theta^-$ :

$$y_j = r_j + \gamma \max_{a'} Q(s_{j+1}, a'; \theta^-) \quad (9)$$

The model went through 80 epochs of training, and we used an experience replay buffer that had 1,500 transitions in it. The batch size was 16. Every 40 steps, we updated the target network parameters. This process helps the model learn and improve over time.

### F. INN-2: Momentum-Augmented DQN

We're making the state more detailed by adding how emotions change from one step to the next:

$$\hat{\mathbf{s}}_t = (v_t, a_t, \Delta v_t, \Delta a_t) \in [-1, 1]^4 \quad (10)$$

This includes how valence and arousal shift, where  $\Delta v_t = v_t - v_{t-1}$  and  $\Delta a_t = a_t - a_{t-1}$ . To do this, we're expanding the input layer of the network to 4 dimensions, using a similar setup to what we had with INN-1.

### G. INN-3: Adaptive Reward Shaping

We replace the fixed penalty with a time-dependent schedule:

$$\lambda(t) = \lambda_{\text{base}} \times \left(1 + \beta \cdot \frac{t}{T}\right) \quad (11)$$

where the base penalty is  $\lambda_{\text{base}} = -100$ , scaled by a factor that increases over the playlist length of 6 steps ( $\beta = 0.8$ ,  $T = 6$ ). This creates a curriculum where early exploration is lenient ( $\lambda(1) \approx -113$ ) and late-stage penalties are severe, reaching  $\lambda(6) = -180$  by the final step.

### H. INN-4: Multi-Target Navigation

Instead of fixing the target to Happy, we train and evaluate the agent for all 8 GEW directions:

$$Q(S_t, A_t, g_k), \quad g_k \in \{g_1, \dots, g_8\} \quad (12)$$

This proves that the DEAP dataset exhibits structural biases toward certain emotions.

### I. Evaluation Protocol

We’re using a special way to test our system, called Leave-One-Subject-Out (LOSO) cross-validation. Here’s how it works: we train our system on 31 people, and then we test it on the 32nd person. We do this 32 times, so each person gets a turn being the one we test on. We’re looking at two main things to see how well our system is working:

- **Convergence rate:** First, we want to know how often our system can get close enough to the target - we consider “close enough” to be within  $< 10^\circ$  angular error from the target.
- **Mean angular error:** Second, we want to know the average angular distance between where our system ends up and where we want it to be - this is called the mean angular error.

When we start, our system is always set to a negative emotion, like being “Anger” (negative valence, high arousal), and we want it to get to a positive emotion, like being “Happy” (positive valence, high arousal). Our system can only use 6 clips to try to get there.

## III. EXPERIMENTAL RESULTS

### A. Baseline Performance

The tabular Q-learning baseline with an 8-action GEW space and fixed reward shaping achieves 68.8% convergence (22/32 subjects), exceeding the 59% (19/32) reported in [2]. The converged group achieves  $1.6^\circ \pm 2.2^\circ$  final error, while the failed group shows  $48.1^\circ \pm 33.2^\circ$ .

Fig. 1 illustrates the effect of reward shaping. Without shaping, the error plateaus after iteration 2–3; with shaping, it decreases across the six-step horizon.

Fig. 2 presents the per-subject trajectory grid across all 32 subjects, with green borders indicating converged subjects and red borders indicating failures.

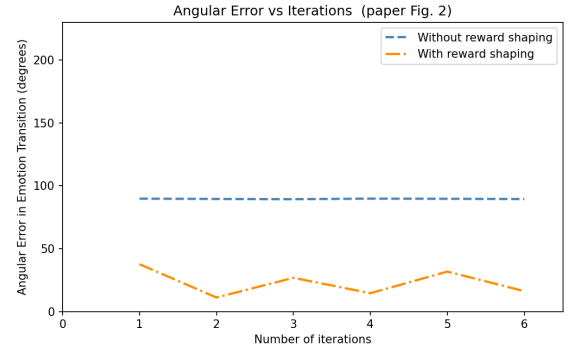


Fig. 1: Baseline angular error over six iterations with and without reward shaping ( $\lambda = -100$ ). Reward shaping reduces stagnation and improves final error.

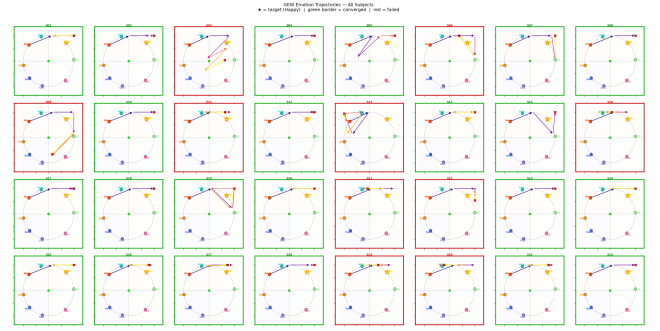


Fig. 2: All-subject trajectory grid for the baseline. Green border = converged ( $< 10^\circ$ ), red border = failed. Star markers indicate the target (Happy).

### B. Comparative Results

Table I presents the comprehensive comparison across all methods. Fig. 3 shows the convergence curves and Fig. 4 presents the convergence rate and mean error side-by-side.

### C. Error Distribution Analysis

Fig. 5 plot of final angular errors. INN-1 the distribution is tight with median around  $0^\circ$  and only 2 outliers. in the baseline bimodal it shows extreme outliers around  $40^\circ$ . INN-2 and INN-3 shows wide interquartile range which is the reason for the inconsistent performance.

### D. Per-Subject Improvement Heatmap

Fig. 6 the figure shows the error improvement of each subject of the each innovation over the baseline. In the green cells the innovation outperforms the baseline bimodal. As we can see INN-1 consistently improves across almost all subjects.

### E. Multi-Target Analysis (INN-4)

Fig. 7 in this the target state is varied across all the 8 states in GEW whole and we can see the convergence rate. In this where the target is ste to positive valance such as Happy the convergence rate (Happy,62.5) is higher but for negative

TABLE I: Comparative Results Across All Methods (LOSO, 32 Subjects)

| Method                  | Conv.        | Rate         | Mean Error                                   |
|-------------------------|--------------|--------------|----------------------------------------------|
| Baseline (Tabular Q)    | 22/32        | 68.8%        | $16.2^\circ \pm 28.5^\circ$                  |
| INN-3: Adaptive Shaping | 17/32        | 53.1%        | $33.2^\circ \pm 49.5^\circ$                  |
| <b>INN-1: DQN (2-D)</b> | <b>30/32</b> | <b>93.8%</b> | <b><math>5.0^\circ \pm 17.3^\circ</math></b> |
| INN-2: Momentum DQN     | 17/32        | 53.1%        | $41.9^\circ \pm 44.3^\circ$                  |

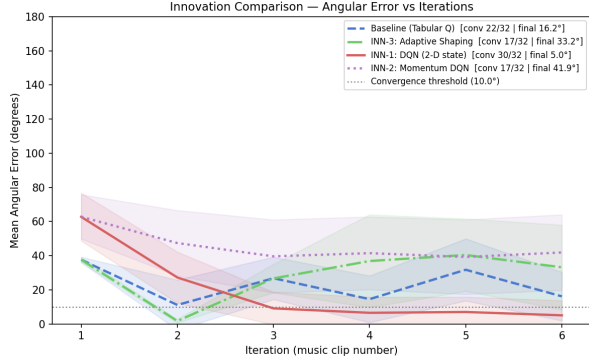


Fig. 3: Mean angular error across the six-step horizon for all methods. INN-1 (DQN) achieves the lowest terminal error.

valance targets such as guilt and Anger the states are rarley reached .this is due to **valence imbalance** in DEAP: under a fixed horizon,basically the clips are biased to transitions to some goal more than other readily

Table II details the per-target results.

#### F. State Coverage Analysis

Fig. 8 shows the comparison of the state visiting frequencies in Baseline and Innovations.In DQN-based Innovations there is more uniform state convergence , which shows better exploration of emotions

### IV. DISCUSSION

#### A. Why Does the Continuous DQN Perform Best?

INN-1 performs best for two main reasons:

- 1) **Elimination of the discretisation bottleneck:** Our 9 state tabular representation considers all positions within a GEW region indential. A listener at  $(v = 0.1, a = 0.8)$  and  $(v = 0.9, a = 0.8)$ .may be assigned same discrete state although there is huge difference in emotional position.. The continous input in the DQN allows neural interpolation in nearby V-A coordinates produce similar Qvalues without explicitlty making it discrete.
- 2) **Synergy with abstract actions:** The 8-action GEW direction space corespond to the more subject-independent genralized policy which work well with the DQN based innovation . The agent can learn more of genralized selection of the music clip and transitions between the emotion state space rather than memorizing clip selection of each individual subject.

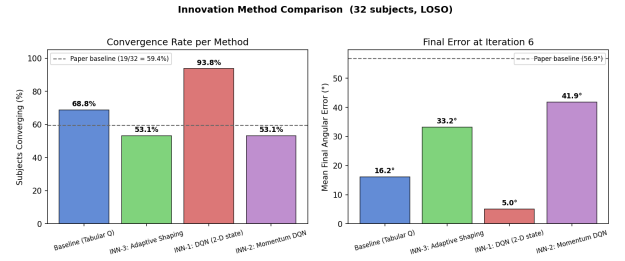


Fig. 4: Convergence rate (%) and mean final angular error (°) across methods. INN-1 achieves the best performance on both metrics.

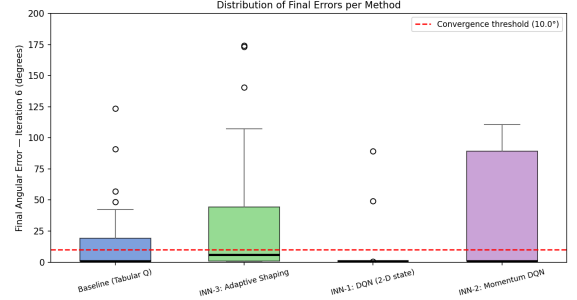


Fig. 5: Box plot of final angular errors. INN-1 achieves the tightest distribution with near-zero median. Baseline is bimodal; INN-2 and INN-3 are inconsistent.

#### B. Why Do INN-2 and INN-3 Underperform?

**INN-2 (Momentum DQN):** In this Innovation we are expanding the stae space from 2D to 4D which increases the complexity and the volume also .Which requires more dat to learn genralized policy. in the given dataset there are only 40 clips per subject due to this there are lesser state-momentum combinations Additionally, computing differences ( $\Delta v$ ,  $\Delta a$ ) expands noise in EEG-derived ratings.

**INN-3 (Adaptive Shaping):** Increase in the penalty can lead to agent selecting the suboptimal clip so basiccally rather from staying in the same state and agents selects clip randomly rather than startegically useful clip slection leading to unstable action selection trajectories. Which is the reason for lower convergence rate 53.1% (vs. 68.8% for the baseline).

#### C. Valence Bias in DEAP

The multi-target analysis (INN-4) shows that the dataset supports some targets more. By observation the targets on the positive valance side have higher convergence rate than the opposite target which are on the negative valance side . This results are very important for interpriting the results on the DEAP dataset it infulence which goals are acchivable under the fixed horizon this is due to the nature of the dataset.

#### D. Limitations

- 1) **Self-report labels:** Valence/arousal values are subjective and it can happen that they does not reflect the actual emotion that the perticular person is feeling now . To

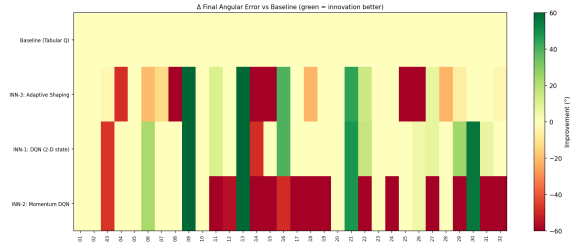


Fig. 6: Per-subject heatmap showing error improvement over the baseline. Green = innovation outperforms baseline.

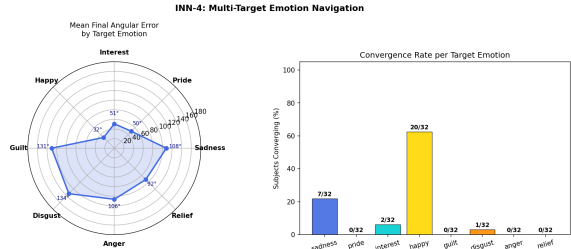


Fig. 7: Multi-target convergence radar (INN-4). Positive-valence targets are generally easier to reach than negative-valence targets in DEAP.

bypass this we would need to process the EEG signals real time and map them to the Emotions.

- 2) **Static response assumption:** In this we are assuming that the clip produces identical emotional reflect on the replay Which may not be the case all the time due to fatigue.
- 3) **Dataset scale:** DEAP provides only 32 subjects and 40 clips. Modern deep RL typically benefits from orders of magnitude more interactions.
- 4) **Offline evaluation:** During the testing period we have used LOSO cross validation .No online validation during the testing is performed.
- 5) **Overestimation bias:**

Satndard DQN can overestimate the Q-values due to maximization bias to overcome this we can use Double DQN in future work

## V. CONCLUSION

We modeled music-driven emotional state navigation as an MDP in the valence-arousal plane and evaluated four extensions to a basic tabular Q-learning approach. The continuous Deep Q-Network (INN-1) **93.8% convergence** (30 out of 32 subjects) with a mean angular error of  $5.0^\circ$ . This corresponds to a 25 percentage-point improvement and a 3.2 times reduction in error compared to the baseline. These improvements come mainly from (i) processing continuous affective states without discretization and (ii) using an 8-direction, subject-independent action space that supports generalization across subjects.

The multi-target evaluation also suggests that DEAP favors certain regulation targets, particularly positive-valence goals,

TABLE II: INN-4 Multi-Target Navigation Results

| Target   | Mean Error    | Conv. Rate    |
|----------|---------------|---------------|
| Happy    | $31.7^\circ$  | 62.5% (20/32) |
| Sadness  | $108.2^\circ$ | 21.9% (7/32)  |
| Interest | $50.8^\circ$  | 6.2% (2/32)   |
| Pride    | $49.9^\circ$  | 0.0% (0/32)   |
| Relief   | $92.3^\circ$  | 0.0% (0/32)   |
| Anger    | $106.3^\circ$ | 0.0% (0/32)   |
| Guilt    | $130.6^\circ$ | 0.0% (0/32)   |
| Disgust  | $133.8^\circ$ | 3.1% (1/32)   |

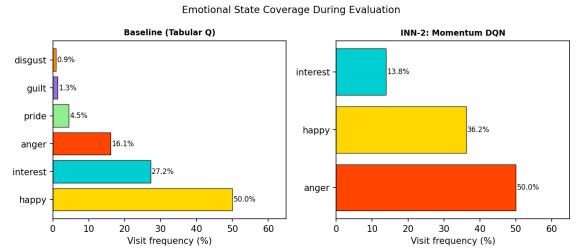


Fig. 8: State visit frequency comparison between baseline and innovations, showing improved exploration with DQN.

over others. This should be taken into account when using the dataset for emotion regulation studies. While momentum augmentation and the proposed adaptive shaping did not enhance performance within our data and horizon limits, these strategies may work better with larger datasets or online interactions.

## A. Future Work

- Online human in loop validation with realtime EEG signals processing
- Transfer learning across larger datasets (EMODB, PMemo).
- Meta-learning for few-shot personalisation to new users.
- Performing Clinical validation of recommended music with music therapy consultants.

## REFERENCES

- [1] World Health Organization, “Depression,” WHO Fact Sheet, 2023. [Online]. Available: <https://www.who.int/news-room/fact-sheets/detail/depression>
- [2] S. Dutta, R. Neogy, and T. Das, “Reinforcement learning for music-driven emotion management,” in *Proc. IEEE Eng. Med. Biol. Conf. (EMBC)*, 2020, pp. 5849–5852.
- [3] S. Koelstra, C. Muhl, M. Soleymani, J.-S. Lee, A. Yazdani, T. Ebrahimi, T. Pun, A. Nijholt, and I. Patras, “DEAP: A database for emotion analysis using physiological signals,” *IEEE Trans. Affect. Comput.*, vol. 3, no. 1, pp. 18–31, 2012.
- [4] J. A. Russell, “A circumplex model of affect,” *J. Pers. Soc. Psychol.*, vol. 39, no. 6, pp. 1161–1178, 1980.
- [5] K. R. Scherer, “What are emotions? And how can they be measured?” *Social Science Information*, vol. 44, no. 4, pp. 695–729, 2005.
- [6] V. Mnih, K. Kavukcuoglu, D. Silver, A. A. Rusu, J. Veness, M. G. Belle-mare, A. Graves, M. Riedmiller, A. K. Fidjeland, G. Ostrovski, *et al.*, “Human-level control through deep reinforcement learning,” *Nature*, vol. 518, no. 7540, pp. 529–533, 2015.
- [7] A. Y. Ng, D. Harada, and S. Russell, “Policy invariance under reward transformations: Theory and application to reward shaping,” in *Proc. Int. Conf. Mach. Learn. (ICML)*, 1999, pp. 278–287.

- [8] H. van Hasselt, A. Guez, and D. Silver, “Deep reinforcement learning with double Q-learning,” in *Proc. AAAI Conf. Artif. Intell.*, 2016, pp. 2094–2100.